Asymptotic Inversion of the Generalized Hyperfactorial K-Function

Lambert and r-Lambert Anchors, Centered Bernoulli Structure, and a General Calculus for Power–Logarithmic Transseries

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ProveIt research note

3 September 2026

Abstract

The generalized hyperfactorial, or K-function, is characterized by $\mathrm{K}(z+1) = z^z\,\mathrm{K}(z)$, $\mathrm{K}(1) = 1$, and admits the exact representations

$$\log \mathrm{K}(z) = \zeta'(-1, z) - \zeta'(-1) = (z-1)\log\Gamma(z) - \log \mathrm{G}(z).$$

This article derives an all-orders asymptotic transseries for the large positive inverse branch $\mathrm{K}_+^{-1}(y)$ as $y \to +\infty$, and then develops a general inversion theory for functions whose logarithms have dominant shape $ax^p(\log x - \beta)$.

The decisive normalization is the midpoint shift $x = q + \frac{1}{2}$. It removes the otherwise present $q\log q$ term and turns the entire algebraic tail into an even series:

$$\log \mathrm{K}\bigl(q + \tfrac{1}{2}\bigr) \sim \frac{q^2}{2}\log q - \frac{q^2}{4} - \frac{1}{24}\log q + \kappa + \sum_{m\geq 1}\frac{c_m}{q^{2m}}, \qquad \kappa = \log A - \frac{1}{8},$$

where A is the Glaisher–Kinkelin constant and

$$c_m = -\frac{B_{2m+2}(1/2)}{(2m)(2m+1)(2m+2)}.$$

For $Y = \log y$, put

$$w = \mathrm{W}_0\left(\frac{4Y}{\mathrm{e}}\right), \qquad Q = 2\sqrt{\frac{Y}{w}} = \exp\left(\frac{w+1}{2}\right), \qquad L = \log Q.$$

Then

$$\mathrm{K}_+^{-1}(y) \sim \frac{1}{2} + Q + \frac{a_1(L)}{Q} + \frac{a_2(L)}{Q^3} + \frac{a_3(L)}{Q^5} + \cdots,$$

with explicit rational coefficient functions, a triangular coefficient recurrence, and an operator formula valid to all orders. A refined anchor absorbs the logarithmic defect into an exactly invertible core. It is expressed through the r-Lambert function with $r = -1/(12\mathrm{e})$, and moves the first correction from order Q^{-1} to order $Q^{-3}/\log Q$.

For the general class

$$F(x) \sim ax^p(\log x - \beta) + \sum_{\lambda\in\Lambda} x^{p-\lambda} r_\lambda(U), \qquad U = p(\log x - \beta),$$

we prove an anchored Lagrange–Bürmann formula, a support-transfer theorem, and a closed all-orders expression for every inverse coefficient. A forward defect of power gap λ produces an inverse defect beginning at power $x^{1-\lambda}$, while mixed interactions are indexed by the additive semigroup generated by the forward gaps. The theory explains, rather than merely observes, the odd inverse-power structure of the K-function. Numerical tests against high-precision Barnes-G evaluation confirm the predicted orders and show that the refined anchor is already highly accurate at moderate arguments.

Keywords. Generalized hyperfactorial; K-function; inverse asymptotics; transseries; Lambert W; r-Lambert function; Hurwitz zeta; Barnes G-function; Bernoulli polynomials; Lagrange–Bürmann inversion.

Research status

Exact functional identities and the real-branch derivative formulas below are classical. The midpoint normal form follows rigorously from the standard Hurwitz-zeta expansion. The inverse coefficient formulas and recurrences are formal consequences of Lagrange–Bürmann inversion and become Poincaré asymptotic expansions on the positive real ray by the remainder estimates in Section 11. Statements about exponentially improved complex expansions are explicitly separated from the algebraic theory.

Contents

1 The problem and the answer at a glance

The discrete hyperfactorial

$$H_n = \prod_{k=1}^{n} k^k$$

has logarithm of order $n^2 \log n$. The analytic continuation relevant here is the K-function, normalized so that

$$\mathrm{K}(n+1) = H_n, \qquad n \in \mathbb{N}. \tag{1}$$

The inverse problem is not to invert K by an elementary operation, but to understand the complete hierarchy of scales in the solution of

$$\mathrm{K}(x) = y, \qquad y \to +\infty. \tag{2}$$

Taking logarithms converts an extremely fast-growing function into one whose dominant part is a quadratic logarithmic monomial. Set

$$Y = \log y. \tag{3}$$

The rough balance $Y \sim \frac{1}{2}x^2 \log x$ already suggests $x \asymp \sqrt{Y/\log Y}$. However, a complete inverse expansion contains three distinct kinds of information:

1. the nested logarithms coming from the ordinary Lambert function;
2. a constant centering correction $1/2$, which is beyond every algebraic order in $1/\log Y$ relative to the leading term;
3. an infinite hierarchy of algebraically small inverse-power sectors.

A single ordinary power series cannot encode all three layers without losing their ordering.

Main result

Let

$$\kappa = \log A - \frac{1}{8}, \qquad w = \mathrm{W}_0\left(\frac{4Y}{\mathrm{e}}\right), \qquad Q = \exp\left(\frac{w+1}{2}\right) = 2\sqrt{\frac{Y}{w}}, \qquad L = \log Q. \tag{4}$$

Define the slow defect

$$\Delta = \frac{L}{24} - \kappa. \tag{5}$$

Then the large positive inverse branch has the Poincaré transseries

$$\boxed{\mathrm{K}_+^{-1}(y) \sim \frac{1}{2} + Q + \sum_{n\ge 1} \frac{a_n(L)}{Q^{2n-1}}.} \tag{6}$$

The first three coefficient functions are

$$a_1(L) = \frac{\Delta}{L}, \tag{7}$$

$$a_2(L) = \frac{7}{5760L} + \frac{\Delta}{24L^2} - \frac{(L+1)\Delta^2}{2L^3}, \tag{8}$$

$$a_3(L) = \frac{(3L^2+5L+3)\Delta^3}{6L^5} - \frac{(4L+3)\Delta^2}{48L^4} + \frac{(1-7L)\Delta}{1920L^3} + \frac{49-186L}{967680L^2}. \tag{9}$$

For every fixed $N \ge 0$,

$$\mathrm{K}_+^{-1}(y) = \frac{1}{2} + Q + \sum_{n=1}^{N} \frac{a_n(L)}{Q^{2n-1}} + \mathcal{O}\left(Q^{-2N-1}\right). \tag{10}$$

All coefficient functions are generated recursively by Theorem 4.2, or directly by the support formula in Theorem 8.2.

There is a second, sharper organization. Let $\mathcal{W}_r$ denote the inverse of $s \mapsto se^s + rs$, and put

$$s = \mathcal{W}_{-1/(12\mathrm{e})}\left(\frac{48(Y-\kappa)+1}{12\mathrm{e}}\right), \qquad Q_* = \exp\left(\frac{s+1}{2}\right), \qquad L_* = \log Q_*. \tag{11}$$

Then

$$\boxed{\mathrm{K}_+^{-1}(y) \sim \frac{1}{2} + Q_* + \sum_{n\geq 1} \frac{b_n(L_*)}{Q_*^{2n+1}},} \tag{12}$$

where

$$b_1(L) = \frac{7}{5760L}, \tag{13}$$

$$b_2(L) = -\frac{186L-49}{967680L^2}, \tag{14}$$

$$b_3(L) = \frac{45720L^2 - 5435L + 637}{464486400L^3}. \tag{15}$$

The refined anchor incorporates the full defect of power gap two; consequently its first unabsorbed forward term has gap four, and the inverse begins at Q_*^{-3}, not Q_*^{-1}.

Scale warning

Expanding only in inverse powers of $\log Y$ gives a correct but incomplete view. The term $1/2$ and every inverse-power sector are beyond all orders of that coarse logarithmic scale. The phrase "the asymptotic expansion" is therefore ambiguous unless the ordered transmonomial scale is stated.

2 The K-function and its large real branch

2.1 Equivalent definitions

For $\Re z > 0$, one convenient definition is [1, 4]

$$\log \mathrm{K}(z) = \zeta'(-1,z) - \zeta'(-1), \tag{16}$$

where $\zeta(s,z)$ is the Hurwitz zeta function and the prime denotes the derivative with respect to s. Equivalent formulas are

$$\mathrm{K}(z) = (2\pi)^{-(z-1)/2} \exp\left\{\binom{z}{2} + \int_0^{z-1} \log\Gamma(t+1)\,\mathrm{d}t\right\}, \tag{17}$$

and, using the Barnes G-function [3, 17],

$$\mathrm{K}(z)\,\mathrm{G}(z) = \Gamma(z)^{z-1}, \tag{18}$$

where G is the Barnes G-function normalized by $\mathrm{G}(z+1) = \Gamma(z)\,\mathrm{G}(z)$ and $\mathrm{G}(1) = 1$.

The Hurwitz shift relation immediately gives

$$\log \mathrm{K}(z+1) - \log \mathrm{K}(z) = z \log z, \tag{19}$$

so that

$$\mathrm{K}(z+1) = z^z\,\mathrm{K}(z), \qquad \mathrm{K}(1) = 1. \tag{20}$$

Iterating (20) proves (1).

2.2 An exact logarithmic derivative

The derivative of (16) is especially useful for inversion. The identities

$$\partial_z\zeta(s,z) = -s\zeta(s+1,z), \qquad \zeta(0,z) = \frac{1}{2} - z, \qquad \zeta'(0,z) = \log\Gamma(z) - \frac{1}{2}\log(2\pi)$$

yield the following formula.

Proposition 2.1 (Exact derivative)**.** *For $x > 0$,*

$$\frac{\mathrm{d}}{\mathrm{d}x}\log \mathrm{K}(x) = \log\Gamma(x) + x - \frac{1}{2} - \frac{1}{2}\log(2\pi), \tag{21}$$

$$\frac{\mathrm{d}^2}{\mathrm{d}x^2}\log \mathrm{K}(x) = \psi(x) + 1, \tag{22}$$

where $\psi = \Gamma'/\Gamma$.

Proof. Differentiate $\partial_z\zeta(s,z) = -s\zeta(s+1,z)$ with respect to s:

$$\partial_z\zeta'(s,z) = -\zeta(s+1,z) - s\zeta'(s+1,z).$$

At $s = -1$, this becomes

$$\partial_z\zeta'(-1,z) = -\zeta(0,z) + \zeta'(0,z),$$

and substitution of the two special values gives (21). A second differentiation gives (22). □

Since $\psi(x) = \log x + O(x^{-1})$, the function $\log \mathrm{K}(x)$ is strictly increasing for all sufficiently large x, and tends to $+\infty$. We use K_+^{-1} for the inverse branch selected by $x \to +\infty$ as $y \to +\infty$.

For orientation, these are the only two positive stationary points. Indeed, $\psi'(x) > 0$, so $\psi(x) + 1$ crosses zero exactly once and $(\log \mathrm{K})'$ has a unique minimum. Moreover, $(\log \mathrm{K})'(x) \to +\infty$ at both endpoints 0^+ and $+\infty$, while $(\log \mathrm{K})'(1) < 0$. The two zeros are numerically

$$x_{\max} = 0.2909569866991807959\ldots, \qquad \mathrm{K}(x_{\max}) = 1.2986723069317708434\ldots,$$
$$x_{\min} = 1.5376888637364865107\ldots, \qquad \mathrm{K}(x_{\min}) = 0.8797868430593994168\ldots.$$

Thus the notation K_+^{-1} is not cosmetic: below the upper extremal value there are additional real local branches. None survives in the limit $y \to +\infty$.

3 Forward asymptotics and the midpoint normal form

3.1 The unshifted expansion

Let A be the Glaisher–Kinkelin constant, characterized by

$$\log A = \frac{1}{12} - \zeta'(-1). \tag{23}$$

The standard large-parameter expansion of the Hurwitz-zeta derivative [4, 12] gives, for every fixed $N \geq 0$,

$$\begin{aligned}\log \mathrm{K}(x) = {}& \left(\frac{x^2}{2} - \frac{x}{2} + \frac{1}{12}\right)\log x - \frac{x^2}{4} + \log A \\ & - \sum_{m=1}^{N} \frac{B_{2m+2}}{(2m)(2m+1)(2m+2)x^{2m}} + \mathcal{O}\big(x^{-2N-2}\big),\end{aligned} \tag{24}$$

initially on the positive real ray, and uniformly in each closed subsector $|\arg x| \le \pi - \delta$ after a branch of $\log x$ is fixed. Here $B_n = B_n(0)$ are Bernoulli numbers. The first terms are

$$\log \mathrm{K}(x) \sim \left(\frac{x^2}{2} - \frac{x}{2} + \frac{1}{12}\right)\log x - \frac{x^2}{4} + \log A + \frac{1}{720x^2} - \frac{1}{5040x^4} + \frac{1}{10080x^6} - \cdots . \tag{25}$$

The asymptotic form (24) is already sufficient for a leading inversion. It is not, however, in the best coordinate. The term $-\frac{1}{2}x\log x$ creates both even and odd inverse powers after reversion. A translation removes it and reveals a hidden Bernoulli symmetry.

3.2 A shifted expansion valid for arbitrary center

The Bernoulli polynomials are defined by

$$\frac{te^{\alpha t}}{e^t - 1} = \sum_{n=0}^{\infty} B_n(\alpha)\frac{t^n}{n!}. \tag{26}$$

They satisfy

$$B_n(\alpha) = \sum_{j=0}^{n} \binom{n}{j} B_j \alpha^{n-j}, \qquad B_n(1-\alpha) = (-1)^n B_n(\alpha). \tag{27}$$

Theorem 3.1 (Shifted K-asymptotics). *Fix $\alpha \in \mathbb{C}$. As $q \to \infty$ in a closed subsector of $|\arg q| < \pi$,*

$$\begin{aligned}\log \mathrm{K}(q+\alpha) \sim \frac{q^2}{2}\log q - \frac{q^2}{4} + (\alpha - \tfrac{1}{2})q\log q + \frac{B_2(\alpha)}{2}\log q \\ + \log A + \frac{\alpha^2 - \alpha}{2} + \sum_{n\ge 1}(-1)^{n+1}\frac{B_{n+2}(\alpha)}{n(n+1)(n+2)q^n}.\end{aligned} \tag{28}$$

After truncation at q^{-N}, the remainder is $O(q^{-N-1})$, with the usual sectorial interpretation when a Stokes direction is approached.

Proof. Insert $x = q + \alpha$ into (24) and use

$$\log(q+\alpha) = \log q + \sum_{j\ge 1} \frac{(-1)^{j+1}}{j}\frac{\alpha^j}{q^j}, \qquad (q+\alpha)^{-2m} = q^{-2m}\sum_{j\ge 0}(-1)^j\binom{2m+j-1}{j}\frac{\alpha^j}{q^j}.$$

At each fixed inverse power only finitely many terms contribute. The polynomial and logarithmic part collects directly into the first line of (28). For $n \ge 1$, the coefficient of q^{-n} is a binomial convolution of Bernoulli numbers with powers of α. Identity (27) reduces that convolution to

$$(-1)^{n+1}\frac{B_{n+2}(\alpha)}{n(n+1)(n+2)}.$$

This proves the formal expansion. Because translation by a fixed α preserves closed proper subsectors and because the expansion (24) has a remainder after every finite truncation, the coefficient collection also preserves the stated Poincaré remainder. $\square$

Remark 3.2. Formula (28) is useful beyond the inverse problem. It packages every translated Euler–Maclaurin coefficient into a single Bernoulli polynomial. In particular, it makes the distinguished role of the midpoint visible without inspecting coefficients one by one.

3.3 The unique logarithmic center and parity collapse

The coefficient of $q \log q$ in (28) vanishes if and only if $\alpha = 1/2$. At the same point, $B_{2m+1}(1/2) = 0$ for all $m \geq 0$, by the reflection identity in (27). Thus the removal of the subleading logarithmic monomial and the parity collapse of the tail occur at exactly the same center.

Corollary 3.3 (Midpoint normal form). *Let*

$$x = q + \frac{1}{2}, \qquad \kappa = \log A - \frac{1}{8}. \tag{29}$$

Then

$$\boxed{\log \mathrm{K}\left(q + \frac{1}{2}\right) \sim \Phi(q) - \frac{1}{24}\log q + \kappa + \sum_{m\geq 1} c_m q^{-2m},} \tag{30}$$

where

$$\Phi(q) = \frac{q^2}{2}\log q - \frac{q^2}{4}, \tag{31}$$

$$c_m = -\frac{B_{2m+2}(1/2)}{(2m)(2m+1)(2m+2)} = -\frac{(2^{-2m-1} - 1)B_{2m+2}}{(2m)(2m+1)(2m+2)}. \tag{32}$$

Explicitly,

$$\begin{aligned} \log \mathrm{K}\left(q + \frac{1}{2}\right) &\sim \frac{q^2}{2}\log q - \frac{q^2}{4} - \frac{1}{24}\log q + \log A - \frac{1}{8} \\ &\quad - \frac{7}{5760q^2} + \frac{31}{161280q^4} - \frac{127}{1290240q^6} + \frac{511}{4866048q^8} - \cdots . \end{aligned} \tag{33}$$

Proof. Set $\alpha = 1/2$ in (28). Since $B_2(1/2) = -1/12$ and $(\alpha^2 - \alpha)/2 = -1/8$, the elementary part has the claimed form. Odd inverse powers vanish because the relevant Bernoulli polynomials have odd degree and are evaluated at the reflection center. For even degree, the multiplication identity gives

$$B_n(1/2) = (2^{1-n} - 1)B_n,$$

which yields (32). □

Reusable method

A translation should be chosen before inversion. For a general forward expansion beginning with

$$ax^p(\log x - \beta) + bx^{p-1}\log x + \cdots,$$

the shift $x = q + \alpha$ changes the coefficient of $q^{p-1}\log q$ to $ap\alpha + b$. The normal-form choice is therefore

$$\alpha = -\frac{b}{ap}.$$

For the K-function, $a = 1/2$, $p = 2$, and $b = -1/2$, giving $\alpha = 1/2$. In problems derived from Euler–Maclaurin expansions, the same center frequently converts ordinary Bernoulli numbers into Bernoulli polynomials at a symmetry point and removes half of the power lattice.

4 The ordinary Lambert anchor

4.1 Exact inversion of the dominant core

We first invert only $\Phi(q)$ from (31). Write

$$\Phi(Q) = Y. \tag{34}$$

If $w = 2\log Q - 1$, then $Q^2 = \mathrm{e}^{w+1}$ and

$$Y = \frac{1}{4} w\mathrm{e}^{w+1}.$$

Hence $w\mathrm{e}^w = 4Y/\mathrm{e}$, which proves the following elementary but central identity.

Proposition 4.1 (Lambert anchor). *For $Y > 0$, the positive solution of $\Phi(Q) = Y$ is*

$$Q = \exp\left(\frac{1 + \mathrm{W}_0(4Y/\mathrm{e})}{2}\right) = 2\sqrt{\frac{Y}{\mathrm{W}_0(4Y/\mathrm{e})}}. \tag{35}$$

If

$$w = \mathrm{W}_0(4Y/\mathrm{e}), \qquad L = \log Q = \frac{w+1}{2}, \tag{36}$$

then

$$\Phi'(Q) = Q\log Q = QL. \tag{37}$$

The entire inverse problem is now a perturbation of an exactly inverted core. Define

$$R(q) = \kappa - \frac{1}{24}\log q + \sum_{m\geq 1} c_m q^{-2m}, \tag{38}$$

so that the centered equation is

$$Y \sim \Phi(q) + R(q). \tag{39}$$

At $q = Q$, the leading perturbation is only $O(\log Q)$, whereas $\Phi'(Q) = Q\log Q$. The first displacement is consequently of order Q^{-1}, and the relative displacement is of order Q^{-2}.

4.2 A normalized implicit equation

Put

$$t = Q^{-2}, \qquad q = QH, \qquad H = 1 + \sum_{n\geq 1} a_n(L)t^n, \qquad w = 2L - 1, \tag{40}$$

and retain $\Delta = L/24 - \kappa$. Multiplying $\Phi(QH) + R(QH) - Y$ by $4/Q^2$ gives the dimensionless equation

$$\begin{aligned}\mathcal{E}(H; t, L) := H^2\big(w + 2\log H\big) - w - 4\Delta t - \frac{t}{6}\log H \\ + 4\sum_{m\geq 1} c_m t^{m+1} H^{-2m} = 0.\end{aligned} \tag{41}$$

This equation contains only nonnegative powers of t when $H = 1 + O(t)$. Moreover,

$$\frac{\partial\mathcal{E}}{\partial H}(1; 0, L) = 4L, \tag{42}$$

which is invertible for large positive L. Coefficient matching is therefore triangular.

Theorem 4.2 (Triangular recurrence). *Let*

$$H_{n-1}(t) = 1 + \sum_{j=1}^{n-1} a_j(L)t^j. \tag{43}$$

Then the coefficients in (6) are determined by

$$\boxed{a_n(L) = -\frac{1}{4L}\,[t^n]\,\mathcal{E}\big(H_{n-1}(t); t, L\big), \qquad n \geq 1.} \tag{44}$$

At order n, only $c_1, \dots, c_{n-1}$ are needed. Each a_n is a rational function of L with coefficients in $\mathbb{Q}[\kappa]$. Moreover,

$$a_n(L) = O(1), \qquad L \to +\infty,$$

so every a_n has a finite limit at infinity.

Proof. Assume $a_1, \dots, a_{n-1}$ have been chosen so that $\mathcal{E}(H_{n-1}; t, L) = O(t^n)$. Replacing H_{n-1} by $H_{n-1} + a_n t^n$ changes the coefficient of t^n by

$$a_n \frac{\partial \mathcal{E}}{\partial H}(1; 0, L) = 4La_n;$$

all nonlinear effects of the new term begin at order t^{n+1}. Solving the linear equation at order t^n gives (44). The assertion about c_m follows from the factor t^{m+1} in (41). Rationality follows inductively from closure of $\mathbb{Q}[\kappa, L, L^{-1}]$ under the finite coefficient operations involved. Finally, $\Delta = O(L)$. If $a_1, \dots, a_{n-1} = O(1)$, then the coefficient extracted in (44) is $O(L)$: the only large factor is the single affine occurrence of $w = 2L - 1$, while all Bernoulli-source terms are bounded. Division by $4L$ gives $a_n = O(1)$. Since a_n is rational in L, boundedness at infinity implies the existence of a finite limit. □

Algorithm 4.3 (Coefficient generation for the ordinary anchor)**.** To compute $a_1, \dots, a_N$:

1. form $c_1, \dots, c_{N-1}$ from (32);
2. initialize $H \leftarrow 1$;
3. for $n = 1, \dots, N$, expand $\mathcal{E}(H; t, L)$ only through t^n, set $a_n = -(4L)^{-1}[t^n]\mathcal{E}$, and replace $H \leftarrow H + a_n t^n$;
4. substitute $t = Q^{-2}$ and add the outer shift $1/2$.

The truncation discipline is important: carrying terms above the current order causes expression swell but contributes nothing to the next coefficient.

4.3 Explicit coefficients

Besides (7)–(9), the next coefficient is

$$\begin{aligned} a_4(L) = &- \frac{(15L^3 + 34L^2 + 35L + 15)\Delta^4}{24L^7} + \frac{(23L^2 + 32L + 15)\Delta^3}{144L^6} \\ &+ \frac{(105L^2 - 13L - 39)\Delta^2}{11520L^5} + \frac{(930L^2 - 206L - 77)\Delta}{967680L^4} \\ &+ \frac{45720L^2 - 5435L + 637}{464486400L^3}. \end{aligned} \tag{45}$$

The limits as $L \to \infty$ begin

$$a_1 \to \frac{1}{24}, \qquad a_2 \to -\frac{1}{1152}, \qquad a_3 \to \frac{1}{27648}, \qquad a_4 \to -\frac{5}{2654208}. \tag{46}$$

These constants arise from repeated nonlinear interaction of the single logarithmic defect $-\frac{1}{24}\log q + \kappa$. The Bernoulli tail begins to contribute to a_2.

Corollary 4.4 (First practical approximants)**.** *With the notation of* (4)*, define*

$$\begin{aligned} X_0(y) &= \frac{1}{2} + Q, \\ X_1(y) &= X_0(y) + \frac{a_1(L)}{Q}, \\ X_2(y) &= X_1(y) + \frac{a_2(L)}{Q^3}, \\ X_3(y) &= X_2(y) + \frac{a_3(L)}{Q^5}. \end{aligned}$$

Then

$$\mathrm{K}_+^{-1}(y) - X_N(y) = \mathcal{O}\big(Q^{-2N-1}\big), \qquad N = 0, 1, 2, 3. \tag{47}$$

5 Anchored Lagrange–Bürmann inversion

The recurrence above is efficient, but a coordinate-free formula explains why it works and extends immediately to other power–logarithmic problems. We use the standard Lagrange–Bürmann theorem [9, 7] in an anchor-adapted coordinate.

Theorem 5.1 (Anchored Lagrange–Bürmann formula)**.** *Let*

$$F(x) = F_0(x) + R(x) \tag{48}$$

be formal, analytic, or asymptotic data in a region where F_0 is locally invertible. Put

$$X = F_0^{-1}(Y), \qquad \mathcal{D}_X = \frac{1}{F_0'(X)}\frac{\mathrm{d}}{\mathrm{d}X}. \tag{49}$$

Whenever the resulting series is meaningful in the chosen formal topology or asymptotic scale, the branch of F^{-1} asymptotic to X is

$$\boxed{F^{-1}(Y) = X + \sum_{n\geq 1} \frac{(-1)^n}{n!} \mathcal{D}_X^{n-1}\left(\frac{R(X)^n}{F_0'(X)}\right).} \tag{50}$$

Proof. Let $\chi = F_0^{-1}$, set $u = F_0(x)$, and define

$$h(u) = R(\chi(u)).$$

The equation $Y = F(x)$ becomes $Y = u + h(u)$. The Lagrange–Bürmann formula for an arbitrary test function φ gives

$$\varphi(u) = \varphi(Y) + \sum_{n\geq 1} \frac{(-1)^n}{n!}\left(\frac{\mathrm{d}}{\mathrm{d}Y}\right)^{n-1}\big(\varphi'(Y)h(Y)^n\big).$$

Choose $\varphi = \chi$. Then

$$\varphi(Y) = X, \qquad \varphi'(Y) = \frac{1}{F_0'(X)}, \qquad h(Y) = R(X), \qquad \frac{\mathrm{d}}{\mathrm{d}Y} = \frac{1}{F_0'(X)}\frac{\mathrm{d}}{\mathrm{d}X} = \mathcal{D}_X.$$

Substitution yields (50). □

The first two correction blocks are worth recording explicitly:

$$F^{-1}(Y) = X - \frac{R}{F_0'} + \frac{RR'}{(F_0')^2} - \frac{F_0'' R^2}{2(F_0')^3} + \cdots, \tag{51}$$

where every function on the right is evaluated at X. This formula is a useful independent check on any coefficient recurrence.

For the ordinary K-anchor,

$$F_0 = \Phi, \qquad F_0'(Q) = Q\log Q, \qquad \mathcal{D}_Q = \frac{1}{Q\log Q}\frac{\mathrm{d}}{\mathrm{d}Q}, \tag{52}$$

and (50) becomes

$$q(Y) \sim Q + \sum_{n\geq 1} \frac{(-1)^n}{n!}\left(\frac{1}{Q\log Q}\frac{\mathrm{d}}{\mathrm{d}Q}\right)^{n-1}\left(\frac{R(Q)^n}{Q\log Q}\right), \tag{53}$$

with R interpreted through its asymptotic expansion (38). Collecting equal powers of Q^{-1} in (53) reproduces (6) and the recurrence (44).

Remark 5.2 (Why anchoring matters). Applying the near-identity reversion formula directly to F is impossible because $F(x) - x$ is not small. The exact inverse of the dominant core first moves the problem into the coordinate $u = F_0(x)$, where the perturbation is small relative to the identity. The operator $\mathcal{D}_X$ is simply $\mathrm{d}/\mathrm{d}u$ written back in the physical coordinate X.

6 A refined r-Lambert anchor

6.1 An exactly invertible affine logarithmic core

The ordinary anchor leaves the slowly varying term $-\frac{1}{24}\log q + \kappa$ to perturbation theory. It can instead be absorbed into the core. The required inverse belongs to the r-Lambert family introduced and studied by Mező and Baricz [10].

Definition 6.1 (The r-Lambert function)**.** For fixed r, let $\mathcal{W}_r(z)$ denote a branch of the inverse of

$$s \longmapsto s\mathrm{e}^s + rs. \tag{54}$$

Thus

$$\mathcal{W}_r(z)\mathrm{e}^{\mathcal{W}_r(z)} + r\mathcal{W}_r(z) = z. \tag{55}$$

When $r = 0$, this reduces to the ordinary Lambert function.

The following general observation is useful for many special-function inversions.

Theorem 6.2 (Exact inversion of an affine logarithmic power core)**.** *Let*

$$\Psi(x) = ax^p \log x + bx^p + c\log x + d, \qquad ap \neq 0. \tag{56}$$

Choose compatible branches and set

$$r = \frac{c}{a}\exp\left(\frac{pb}{a}\right), \tag{57}$$

$$z = \frac{p}{a}\exp\left(\frac{pb}{a}\right)\left(Y - d + \frac{cb}{a}\right). \tag{58}$$

If $s = \mathcal{W}_r(z)$, then a solution of $\Psi(x) = Y$ is

$$\boxed{x = \exp\left(\frac{s}{p} - \frac{b}{a}\right).} \tag{59}$$

Every local inverse branch is obtained by compatible choices of the logarithm and of a branch of $\mathcal{W}_r$.

Proof. Put

$$s = p\log x + \frac{pb}{a}.$$

Then $\log x = s/p - b/a$ and $x^p = \exp(s - pb/a)$. The two power terms collapse:

$$ax^p \log x + bx^p = \frac{a}{p}\exp\left(-\frac{pb}{a}\right)s\mathrm{e}^s.$$

The remaining logarithmic term is $cs/p - cb/a$. Therefore $\Psi(x) = Y$ is equivalent to

$$s\mathrm{e}^s + \frac{c}{a}\exp\left(\frac{pb}{a}\right)s = \frac{p}{a}\exp\left(\frac{pb}{a}\right)\left(Y - d + \frac{cb}{a}\right),$$

which is precisely (55) with the data (57)–(58). □

6.2 Application to the K-function

Define the refined core

$$\Psi(q) = \frac{q^2}{2}\log q - \frac{q^2}{4} - \frac{1}{24}\log q + \kappa. \tag{60}$$

In Theorem 6.2, take

$$a = \frac{1}{2}, \qquad p = 2, \qquad b = -\frac{1}{4}, \qquad c = -\frac{1}{24}, \qquad d = \kappa.$$

Then

$$r = -\frac{1}{12\mathrm{e}}, \qquad z = \frac{48(Y-\kappa)+1}{12\mathrm{e}}. \tag{61}$$

The positive large solution is exactly Q_* from (11). Its derivative is

$$\Psi'(q) = q\log q - \frac{1}{24q}. \tag{62}$$

The residual perturbation is now only

$$S(q) = \sum_{m\geq 1} c_m q^{-2m}. \tag{63}$$

Thus the first inverse displacement is

$$-\frac{S(Q_*)}{\Psi'(Q_*)} = \frac{7}{5760 L_* Q_*^3} + O(Q_*^{-5}L_*^{-1}),$$

which explains the improved scale before any detailed calculation.

6.3 Refined recurrence and coefficients

Let

$$t = Q_*^{-2}, \qquad q = Q_* H, \qquad H = 1 + \sum_{n\geq 1} b_n(L_*)t^{n+1}, \qquad w = 2L_* - 1. \tag{64}$$

The normalized equation is

$$\begin{aligned}\mathcal{E}_*(H;t,L_*) := H^2\big(w + 2\log H\big) - w - \frac{t}{6}\log H \\ + 4\sum_{m\geq 1} c_m t^{m+1} H^{-2m} = 0.\end{aligned} \tag{65}$$

The term $-4\Delta t$ has disappeared because it belongs to the exact core.

Theorem 6.3 (Refined triangular recurrence)**.** *Let*

$$H^*_{n-1}(t) = 1 + \sum_{j=1}^{n-1} b_j(L)t^{j+1}. \tag{66}$$

Then

$$\boxed{b_n(L) = -\frac{1}{4L}\left[t^{n+1}\right]\mathcal{E}_*\big(H^*_{n-1}(t);t,L\big), \qquad n\geq 1.} \tag{67}$$

Each $b_n(L)$ is rational in L, satisfies $b_n(L) = O(L^{-1})$, and uses only $c_1,\dots,c_n$.

Proof. The proof is identical to that of Theorem 4.2; the linearization remains $4L$. The source term at the first unmatched order has no factor growing faster than a fixed power of L^0, while division by the linearization contributes L^{-1}. Induction gives $b_n = O(L^{-1})$. □

In addition to (13)–(15), one finds

$$
b_4(L) = -\frac{12877200L^3 - 703428L^2 + 57552L + 539}{122624409600L^4}, \tag{68}
$$

$$
b_5(L) = \frac{1539856241280L^4 - 45140228640L^3 + 1666058823L^2 - 38145107L - 25162137}{8034351316992000L^5}. \tag{69}
$$

Their leading large-L behavior recovers the negatives of the forward tail coefficients divided by L:

$$
Lb_1 \to \frac{7}{5760}, \quad Lb_2 \to -\frac{31}{161280}, \quad Lb_3 \to \frac{127}{1290240}, \quad Lb_4 \to -\frac{511}{4866048}. \tag{70}
$$

This is exactly what the linear approximation $-S(Q_*)/\Psi'(Q_*)$ predicts.

The anchored operator form is

$$
q(Y) \sim Q_* + \sum_{n\geq 1} \frac{(-1)^n}{n!} \left(\frac{1}{\Psi'(Q_*)} \frac{\mathrm{d}}{\mathrm{d}Q_*}\right)^{n-1} \left(\frac{S(Q_*)^n}{\Psi'(Q_*)}\right). \tag{71}
$$

For every fixed N,

$$
\mathrm{K}_+^{-1}(y) = \frac{1}{2} + Q_* + \sum_{n=1}^{N} \frac{b_n(L_*)}{Q_*^{2n+1}} + \mathcal{O}\left(\frac{Q_*^{-2N-3}}{L_*}\right). \tag{72}
$$

Summary

The ordinary anchor is simpler because it uses the ubiquitous W_0 and already gives excellent approximations. The refined anchor is structurally cleaner: it absorbs every term of forward power gap two, leaves only the Bernoulli tail, and gives a remainder smaller by two powers of the anchor. The price is evaluation of a single r-Lambert function, which itself is easily computed by Newton iteration.

7 Expansion in elementary logarithms

The Lambert anchors are the most compact and numerically stable form of the answer. For comparison with conventional asymptotic notation, one may expand them further into nested logarithms. This operation should be performed only after the inverse has been organized into its distinct transseries sectors.

7.1 Expansion of the ordinary anchor

Set

$$
M = \log\left(\frac{4Y}{\mathrm{e}}\right), \qquad N = \log M. \tag{73}
$$

The classical principal-branch expansion [2] is

$$
\begin{aligned}
\mathrm{W}_0(4Y/\mathrm{e}) \sim M - N + \frac{N}{M} + \frac{N(N-2)}{2M^2} + \frac{N(2N^2-9N+6)}{6M^3} \\
+ \frac{N(3N^3-22N^2+36N-12)}{12M^4} + \cdots.
\end{aligned} \tag{74}
$$

Substitution into $Q = 2\sqrt{Y/\mathrm{W}_0(4Y/\mathrm{e})}$ yields

$$
\begin{aligned}
Q \sim 2\sqrt{\frac{Y}{M}}\Bigg[1 + \frac{N}{2M} + \frac{N(3N-4)}{8M^2} + \frac{N(5N^2-16N+8)}{16M^3} \\
+ \frac{N(105N^3-568N^2+720N-192)}{384M^4} + \cdots\Bigg].
\end{aligned} \tag{75}
$$

The polynomials in N can be generated to arbitrary order either from the standard Lambert polynomials or by substituting an ansatz in $w + \log w = M$.

The first inverse-power sector begins with

$$\frac{a_1(L)}{Q} = \sqrt{\frac{M}{Y}}\left[\frac{1}{48} - \frac{\kappa + N/96}{M} + \mathcal{O}\left(\frac{N^2}{M^2}\right)\right]. \tag{76}$$

Combining (75) and (76) gives the first three sectors explicitly:

$$\begin{aligned} \mathrm{K}_+^{-1}(y) = 2\sqrt{\frac{Y}{M}}&\left[1 + \frac{N}{2M} + \frac{N(3N-4)}{8M^2} + \frac{N(5N^2 - 16N + 8)}{16M^3} + \mathcal{O}\left(\frac{N^4}{M^4}\right)\right] \\ &+ \frac{1}{2} + \sqrt{\frac{M}{Y}}\left[\frac{1}{48} - \frac{\kappa + N/96}{M} + \mathcal{O}\left(\frac{N^2}{M^2}\right)\right] + \mathcal{O}\left(\frac{M^{3/2}}{Y^{3/2}}\right). \end{aligned} \tag{77}$$

7.2 Why the constant is beyond all logarithmic orders

The leading sector has size $Q \asymp \sqrt{Y/M}$. Relative to it, the centering term has size

$$\frac{1/2}{Q} \asymp \sqrt{\frac{M}{Y}}.$$

Since $Y = (\mathrm{e}/4)e^M$, this ratio is exponentially small as a function of the coarse variable M. Hence no finite or infinite Poincaré series in powers of $1/M$ multiplying the leading scale can produce the constant $1/2$. The same is true, a fortiori, for the inverse-power sectors.

Scale warning

The statement

$$\mathrm{K}_+^{-1}(y) \sim 2\sqrt{Y/M}\left(1 + \frac{N}{2M} + \cdots\right)$$

is correct in the one-sector logarithmic sense, but it does not determine the additive constant and therefore cannot be differentiated, composed, or used as a high-accuracy numerical formula beyond its declared scale. The Lambert- anchored transseries (6) is the more informative object.

7.3 The transmonomial grid

Expanding the ordinary anchor and every coefficient function produces a grid whose typical monomials are

$$Y^{1/2-n} M^{n-1/2-j} N^k, \qquad n, j \in \mathbb{N}_0, \tag{78}$$

with a degree bound on k depending linearly on j, together with the separate constant sector $1/2$. The ordering is lexicographic by exponential size first:

$$\sqrt{\frac{Y}{M}} \succ 1 \succ \sqrt{\frac{M}{Y}} \succ \left(\frac{M}{Y}\right)^{3/2} \succ \cdots,$$

and inside each sector by descending powers of M, with powers of $N = \log M$ treated as coefficients of slower growth.

This grid is not arbitrary. In the centered forward expansion, the set of power gaps is $\{2, 4, 6, \ldots\}$. The general support-transfer theorem in Theorem 8.2 maps a total forward gap $2n$ to inverse power Q^{1-2n}, exactly the odd inverse-power lattice in (6).

8 General inversion of power–logarithmic transseries

We now isolate the structure that does not depend on the special coefficients of the K-function.

8.1 The universal Lambert anchor

Consider the dominant model

$$F_0(x) = ax^p(\log x - \beta), \qquad a \neq 0, \quad p > 0. \tag{79}$$

For the positive real problem assume $a > 0$ and take $Y > 0$ large. Define

$$U = \mathrm{W}_0\left(\frac{p}{a}e^{-p\beta}Y\right). \tag{80}$$

Then

$$X = \exp\left(\beta + \frac{U}{p}\right) = \left(\frac{pY}{aU}\right)^{1/p} \tag{81}$$

solves $F_0(X) = Y$. Indeed,

$$U = p(\log X - \beta), \qquad F_0'(X) = aX^{p-1}(U+1). \tag{82}$$

Theorem 8.1 (Universal power–log anchor). *Every inverse problem with dominant term* (79) *admits an exact Lambert anchor* (81)*. On a complex sector, replace* W_0 *by a compatible branch* W_k *and use the chosen logarithm in* $X = \exp(\beta + U/p)$*. All algebraic corrections can then be generated by the anchored formula* (50).

The examples in Table 1 show that the same core occurs in many classical inversions.

Table 1: Typical power–logarithmic anchors. Lower-order terms determine the subsequent transseries but not the leading Lambert coordinate.

Problem	p	a	β	Dominant logarithm
Inverse Γ	1	1	1	$\log\Gamma(x) \sim x(\log x - 1)$
Inverse Barnes G	2	$1/2$	$3/2$	$\log G(x+1) \sim \frac{1}{2}x^2(\log x - \frac{3}{2})$
Inverse centered K	2	$1/2$	$1/2$	$\log K(q+1/2) \sim \frac{1}{2}q^2(\log q - \frac{1}{2})$
Generalized hyperfactorial with exponent k^r	$r+1$	$1/(r+1)$	$1/(r+1)$	$\sum_{k\le n} k^r \log k \sim \frac{n^{r+1}}{r+1}(\log n - \frac{1}{r+1})$

8.2 The anchor differential operator

Let $U = p(\log X - \beta)$. The operator in (49) takes the simple form

$$\mathcal{D}_X = \frac{X^{-p}}{a(U+1)}\left(X\frac{\mathrm{d}}{\mathrm{d}X}\right). \tag{83}$$

For any exponent μ and coefficient function $f(U)$,

$$\mathcal{D}_X\big(X^\mu f(U)\big) = X^{\mu-p}\mathfrak{T}_\mu f(U), \tag{84}$$

where

$$\mathfrak{T}_\mu f := \frac{\mu f + pf'}{a(U+1)}. \tag{85}$$

Thus every application of $\mathcal{D}_X$ lowers the power of X by exactly p, while acting on the slow coefficient through a first-order rational differential operator. This separation of fast and slow variables is the key to an explicit support theorem.

8.3 Grid-based perturbations

Let $\Lambda \subset (0, \infty)$ be a locally finite set of positive power gaps. Consider

$$F(x) \sim F_0(x) + R(x), \qquad R(x) \sim \sum_{\lambda \in \Lambda} x^{p-\lambda} r_\lambda(U), \qquad U = p(\log x - \beta). \tag{86}$$

The coefficient functions may be rational functions of U, polynomials in $\log U$ with rational-function coefficients, or elements of any differential algebra $\mathcal{A}$ closed under $f \mapsto f'$ and division by $U + 1$. Write $\langle \Lambda \rangle$ for the additive semigroup generated by Λ.

For $m \geq 0$, define the iterated operator

$$\mathfrak{T}_\mu^{\langle m \rangle} = \begin{cases} \mathrm{id}, & m = 0, \\ \mathfrak{T}_{\mu-(m-1)p} \circ \cdots \circ \mathfrak{T}_{\mu-p} \circ \mathfrak{T}_\mu, & m \geq 1. \end{cases} \tag{87}$$

Theorem 8.2 (Support transfer and all-orders coefficients)**.** *Assume that every $\sigma \in \langle \Lambda \rangle$ has only finitely many ordered representations $\sigma = \lambda_1 + \cdots + \lambda_n$ with $\lambda_j \in \Lambda$. Then the inverse branch asymptotic to the Lambert anchor X has the transseries*

$$\boxed{F^{-1}(Y) \sim X + \sum_{\sigma \in \langle \Lambda \rangle} X^{1-\sigma} C_\sigma(U),} \tag{88}$$

where

$$C_\sigma(U) = \sum_{n \geq 1} \frac{(-1)^n}{n!} \sum_{\substack{\lambda_1, \ldots, \lambda_n \in \Lambda \\ \lambda_1 + \cdots + \lambda_n = \sigma}} \mathfrak{T}^{\langle n-1 \rangle}_{(n-1)p+1-\sigma} \left(\frac{\prod_{j=1}^n r_{\lambda_j}(U)}{a(U+1)} \right). \tag{89}$$

For each fixed σ, both sums are finite. In particular, a forward block with gap λ produces a first-order inverse block

$$-X^{1-\lambda} \frac{r_\lambda(U)}{a(U+1)}, \tag{90}$$

and an n-fold interaction with total gap σ always has fast power $X^{1-\sigma}$.

Proof. Expand the n-th summand of (50). An ordered product of forward blocks indexed by $\lambda_1, \ldots, \lambda_n$ contributes

$$R(X)^n \supset X^{np-\sigma} \prod_{j=1}^n r_{\lambda_j}(U), \qquad \sigma = \sum_{j=1}^n \lambda_j.$$

After division by $F_0'(X) = aX^{p-1}(U+1)$, its power is

$$\mu_0 = (n-1)p + 1 - \sigma.$$

Each of the remaining $n - 1$ applications of $\mathcal{D}_X$ lowers the power by p. Formula (84) therefore gives the final power

$$\mu_0 - (n-1)p = 1 - \sigma$$

and the slow coefficient

$$\mathfrak{T}^{\langle n-1 \rangle}_{\mu_0} \left(\frac{\prod_j r_{\lambda_j}}{a(U+1)} \right).$$

Multiplication by $(-1)^n/n!$ and summation over all ordered decompositions of σ gives (89). Local finiteness of the support makes coefficient extraction legitimate. □

Corollary 8.3 (Power-gap law)**.** *If the smallest forward gap is $\lambda_0 > 0$, then the relative correction to the anchor begins at $X^{-\lambda_0}$. The n-fold self-interaction of that smallest block begins at relative order $X^{-n\lambda_0}$. Mixed interactions add their gaps.*

Proof. Divide every term in (88) by X. The relative power corresponding to total gap σ is $X^{-\sigma}$. □

8.4 The K-function as a support theorem example

For the ordinary centered K-anchor,

$$p = 2, \qquad a = \frac{1}{2}, \qquad \beta = \frac{1}{2}, \qquad U = 2\log Q - 1. \tag{91}$$

The perturbation (38) can be written as

$$R(Q) = Q^{2-2} r_2(U) + \sum_{m\geq 1} Q^{2-(2m+2)} r_{2m+2}(U), \tag{92}$$

with

$$r_2(U) = \kappa - \frac{U+1}{48}, \qquad r_{2m+2}(U) = c_m. \tag{93}$$

Thus $\Lambda = \{2, 4, 6, \ldots\}$, and $\langle\Lambda\rangle = \{2, 4, 6, \ldots\}$. The support-transfer theorem gives powers Q^{1-2n}, precisely the odd inverse powers in (6). It also gives, without recurrence,

$$a_n(L) = C_{2n}(2L-1). \tag{94}$$

In the refined organization, the entire gap-two block is absorbed into Ψ. The remaining gaps are $\{4, 6, 8, \ldots\}$, so the first inverse power is $Q_*^{1-4} = Q_*^{-3}$. This is the structural reason for the gain in (12).

Reusable method

The support-transfer theorem is a transseries analogue of a Newton polygon. Before computing a single coefficient, it predicts:

1. which inverse powers can occur;
2. where each forward block first appears in the inverse;
3. which mixed products can contribute at a requested order;
4. which terms may be absorbed into a better exact anchor.

This information is especially valuable in computer algebra, where a naive univariate expansion often generates many coefficients that must later cancel.

8.5 A normalized equation for arbitrary power gaps

The closed coefficient formula (89) is conceptually useful, but direct coefficient matching is usually faster. Write

$$x = XH, \qquad H = 1 + \sum_{\sigma\in\langle\Lambda\rangle} h_\sigma(U) X^{-\sigma}. \tag{95}$$

Since $p(\log(XH) - \beta) = U + p\log H$, multiplication of $F(XH) - F_0(X)$ by $p/(aX^p)$ gives

$$\begin{aligned}\mathcal{E}_{p,a}(H) := {} & H^p\big(U + p\log H\big) - U \\ & + \frac{p}{a}\sum_{\lambda\in\Lambda} X^{-\lambda} H^{p-\lambda} r_\lambda\big(U + p\log H\big) = 0.\end{aligned} \tag{96}$$

Its linearization is

$$\left.\frac{\partial\mathcal{E}_{p,a}}{\partial H}\right|_{H=1, X^{-1}=0} = p(U+1). \tag{97}$$

Theorem 8.4 (General triangular recurrence). *Order $\langle\Lambda\rangle$ increasingly. Suppose all h_τ with $\tau < \sigma$ have been determined, and put*

$$H_{<\sigma} = 1 + \sum_{\tau<\sigma} h_\tau(U) X^{-\tau}.$$

Then

$$\boxed{h_\sigma(U) = -\frac{1}{p(U+1)}[X^{-\sigma}]\mathcal{E}_{p,a}(H_{<\sigma}).} \tag{98}$$

Moreover $h_\sigma = C_\sigma$, where C_σ is given by (89).

Proof. The coefficient $h_\sigma X^{-\sigma}$ enters the normalized equation linearly at its first occurrence, with multiplier $p(U+1)$ by (97). Every nonlinear expression containing the new coefficient has total gap greater than σ. This proves the recurrence. Both the recurrence and the anchored Lagrange formula construct the unique formal solution with $H = 1 + o(1)$, so their coefficients agree. □

For a one-dimensional grid $\Lambda \subset d\mathbb{N}$, set $t = X^{-d}$; then (98) reduces to ordinary power-series coefficient matching. The K-function has $d = 2$, which is why the recurrences (44) and (67) are especially simple.

8.6 Coefficient algebras and nested logarithms

The slow variable U is itself a Lambert function of Y. Several useful coefficient algebras are stable under the operator $\mathfrak{T}_\mu$:

$$\begin{aligned}&\mathbb{C}(U),\\ &\mathbb{C}(U)[\log U],\\ &\mathbb{C}((U^{-1}))[\log U],\\ &\mathbb{C}((U^{-1}))((\log U)^{-1}),\end{aligned}$$

with the last expression understood as an iterated formal field when needed. The reason is elementary: differentiation preserves these classes and $1/(U+1)$ has a geometric expansion in U^{-1}. Thus a forward polynomial–logarithmic coefficient system remains polynomial–logarithmic after inversion.

If a coefficient contains deeper iterated logarithms, for example $\log\log x$, rewrite it as a function of

$$U, \qquad \log U, \qquad \log\log U, \quad \ldots.$$

The chain rule introduces only rational combinations of these slow variables. The support theorem is unchanged because it tracks the fast powers separately from the coefficient algebra.

Proposition 8.5 (Closure under inversion). *Let $\mathcal{A}$ be a differential algebra of germs or formal series in U that is closed under multiplication and under division by $U+1$. If every $r_\lambda \in \mathcal{A}$, then every inverse coefficient $C_\sigma \in \mathcal{A}$.*

Proof. Formula (89) uses only finite sums, products, derivatives, and division by $a(U+1)$. These operations preserve $\mathcal{A}$. □

8.7 From grids to well-based transseries

The finite-decomposition hypothesis in Theorem 8.2 is automatic for a finitely generated positive grid. It can be weakened in the usual Hahn- and transseries setting [16, 5]. Let the power gaps take values in an ordered abelian group and suppose their support is well-ordered in the direction of increasing smallness. Neumann's lemma then ensures that finite sums of support elements remain well-ordered and that each coefficient in products receives only finitely many contributions under the standard Hahn-series hypotheses. The proof of Theorem 8.2 carries over verbatim.

What must not be dropped is some form of local finiteness. A set of positive gaps accumulating at zero can destroy the existence of a smallest correction and make the formal implicit equation non-triangular. In analytic applications this signals that the chosen anchor has not separated the scales sufficiently; a continuous Mellin spectrum or an additional transmonomial may be required.

8.8 Mixed exponential sectors

Suppose the forward transseries also contains exponentially small blocks,

$$R_{\exp}(x) = \sum_{\omega\in\Omega} e^{-\omega x^{\rho}} x^{\nu_\omega} r_\omega(U). \tag{99}$$

The anchored formula remains valid in a transseries field containing these monomials. At first order, each exponential block transfers to

$$\delta x_\omega \sim -\frac{e^{-\omega X^{\rho}} X^{\nu_\omega} r_\omega(U)}{aX^{p-1}(U+1)}. \tag{100}$$

Products of forward exponentials add their actions ωX^{ρ}, while algebraic gaps add exactly as in Theorem 8.2. This observation is the algebraic core of the Stokes-transfer discussion in Section 13.

9 Normal forms and anchor design

The quality of an inverse transseries depends less on the number of terms than on the choice of the core that is inverted exactly. This section systematizes that choice.

9.1 Translation normal form

Assume

$$F(x) = ax^{p}(\log x - \beta) + bx^{p-1}\log x + cx^{p-1} + O(x^{p-2}\log^{d} x). \tag{101}$$

Set $x = q + \alpha$. The coefficient of $q^{p-1}\log q$ becomes $ap\alpha + b$. Hence

$$\alpha_* = -\frac{b}{ap} \tag{102}$$

removes the largest untranslated defect. The remaining coefficient of q^{p-1} may then be absorbed into a redefinition of β, if desired. For Euler–Maclaurin expansions, α_* often coincides with a symmetry point of the relevant Bernoulli polynomials.

9.2 Absorbing affine logarithmic defects

After translation, one often obtains

$$F(q) = aq^{p}\log q + bq^{p} + c\log q + d + S(q). \tag{103}$$

Theorem 6.2 shows that the first four terms are not merely "approximately invertible": they have an exact inverse in terms of $\mathcal{W}_r$. If $S(q) = O(q^{-\eta})$, this raises the first inverse correction from the scale created by $c\log q + d$ to $O(q^{1-p-\eta}/\log q)$.

For the K-function, the ordinary Lambert core leaves a gap-two defect, whereas the affine core leaves a gap-four defect. This is why the refined anchor gains two powers of Q.

9.3 A hierarchy of possible anchors

There is no absolute "best" anchor. Three levels are useful:

1. **Elementary power–log anchor.** Use $ax^p(\log x - \beta)$ and ordinary W. This maximizes portability and usually gives a closed all-orders coefficient grid.
2. **Affine logarithmic anchor.** Add $c \log x + d$ and use $\mathcal{W}_r$. This often removes the leading inverse sector.
3. **Numerical implicit anchor.** Absorb several algebraic tail terms into a monotone core and solve it numerically. The residual transseries becomes shorter, but the anchor is no longer a named closed-form function.

The first two retain symbolic transparency. The third can be optimal for high-precision numerical work.

Reusable method

Choose the anchor by a cost model, not by aesthetics. A useful objective is to minimize

$$\frac{|R(X)|}{|XF_0'(X)|},$$

the predicted relative first correction, subject to the cost and branch complexity of evaluating F_0^{-1}. The ordinary and refined K-anchors are two Pareto-optimal points in this tradeoff.

10 Algorithms for symbolic and numerical inversion

10.1 Four complementary algorithms

The general theory supports four computational strategies.

Triangular coefficient matching. Use (96) and (98). This is normally the fastest way to obtain a long symbolic expansion because every order is solved by one linear division.

Anchored Lagrange–Bürmann blocks. Use (50) or (89). This gives a closed formula, proves support constraints, and is excellent for checking a recurrence. Repeated differentiation can cause expression swell at high orders.

Asymptotic fixed-point iteration. Rewrite

$$x = F_0^{-1}(Y - R(x))$$

and iterate from $x_0 = X$. Truncating after every operation makes this an efficient formal algorithm. Without truncation, intermediate expressions usually grow unnecessarily.

Newton polishing. Given any asymptotic approximation x_0, set

$$x_{j+1} = x_j - \frac{\log \mathrm{K}(x_j) - Y}{\log \Gamma(x_j) + x_j - \frac{1}{2} - \frac{1}{2}\log(2\pi)}. \tag{104}$$

One or two iterations recover nearly all available working precision when the Lambert anchor is already in the asymptotic regime.

10.2 A sparse coefficient data structure

For a general grid, represent a term by a pair $(\sigma, f(U))$, standing for $X^{-\sigma} f(U)$ in the relative factor H. Store terms in a map keyed by σ. Multiplication adds keys; differentiation applies (84); truncation discards keys above the requested total gap. This representation prevents accidental expansion of $U = \mathrm{W}(cY)$ into nested logarithms before the final presentation stage.

Algorithm 10.1 (Sparse transseries reversion)**.** Given p, a, β, a gap set Λ, coefficient functions r_λ, and a cutoff Σ:

1. form the exact anchor U and X from (80)–(81);
2. initialize the sparse relative series $H = 1$;
3. enumerate $\sigma \in \langle \Lambda \rangle$ in increasing order up to Σ;
4. compute only the coefficient of $X^{-\sigma}$ in (96);
5. divide by $-p(U+1)$ and insert the resulting h_σ into H;
6. return $x = XH$, optionally expanding U into elementary logarithms.

10.3 Residual-driven adaptive truncation

A practical implementation need not predict the optimal number of asymptotic terms. Compute successive truncations x_N, evaluate the residual

$$E_N = F(x_N) - Y, \tag{105}$$

and stop when $|E_N / F'(x_N)|$ ceases to decrease or falls below the target absolute error. This criterion works for convergent local inverse series and for divergent Poincaré series truncated near their least term.

11 Analytic validity and error transfer

Formal transseries become useful analytic statements only after residuals are controlled. For the positive real K-branch, this step is particularly simple because the logarithmic derivative is explicit and positive in the large-argument region.

11.1 A residual certificate

Lemma 11.1 (Real residual bound)**.** *Let F be continuously differentiable and strictly increasing on an interval I. Let $x \in I$ solve $F(x) = Y$, and let $\widehat{x} \in I$. If*

$$m = \inf_{u \in I} F'(u) > 0, \tag{106}$$

then

$$|x - \widehat{x}| \le \frac{|F(\widehat{x}) - Y|}{m}. \tag{107}$$

If F'' is controlled and the residual is small, then more precisely

$$x - \widehat{x} = -\frac{F(\widehat{x}) - Y}{F'(\widehat{x})} + \mathcal{O}\left(\frac{F''(\widehat{x})[F(\widehat{x}) - Y]^2}{F'(\widehat{x})^3} \right). \tag{108}$$

Proof. The mean value theorem gives

$$F(\widehat{x}) - F(x) = F'(\xi)(\widehat{x} - x)$$

for some ξ between x and $\widehat{x}$, proving (107). Taylor expansion about $\widehat{x}$, followed by one formal solution step for $x - \widehat{x}$, gives (108). □

For the K-problem we apply this lemma to $F(x) = \log \mathrm{K}(x)$. By (21),

$$F'(x) = \log \Gamma(x) + x - \frac{1}{2} - \frac{1}{2}\log(2\pi) \sim x \log x. \tag{109}$$

Therefore a forward residual of size E produces an inverse error of size approximately $E/(x \log x)$.

11.2 General error-transfer principle

Proposition 11.2 (Transfer of a forward remainder)**.** *Let $F_0(x) = ax^p(\log x - \beta)$ with $a > 0$, and let $X = F_0^{-1}(Y)$. Suppose a forward term or remainder has size*

$$E(X) = \mathcal{O}\Big(X^{p-\rho}L^d\Big), \qquad L = \log X, \qquad \rho > 0. \tag{110}$$

Then its first-order effect on the inverse has size

$$\delta X = \mathcal{O}\Big(X^{1-\rho}L^{d-1}\Big). \tag{111}$$

The relative inverse error is $O(X^{-\rho}L^{d-1})$.

Proof. The derivative of the core is $F_0'(X) = aX^{p-1}(p(\log X - \beta) + 1) \asymp X^{p-1}L$. Use the first anchored correction $-E(X)/F_0'(X)$. □

This proposition is the analytic counterpart of the formal power-gap law. It also shows why logarithmic defects are mild: one factor of $\log X$ is lost when a forward residual is divided by the derivative.

11.3 Remainder of the ordinary K-expansion

Let

$$H_N = 1 + \sum_{n=1}^{N} a_n(L)t^n, \qquad t = Q^{-2}. \tag{112}$$

The recurrence makes the normalized residual

$$\mathcal{E}(H_N; t, L) = O(Lt^{N+1}). \tag{113}$$

Multiplying back by $Q^2/4$ gives

$$\log \mathrm{K}\left(\frac{1}{2} + QH_N\right) - Y = O(LQ^{-2N}). \tag{114}$$

Since the derivative is asymptotic to QL, Theorem 11.1 gives

$$q - QH_N = O(Q^{-2N-1}), \tag{115}$$

which proves (10). The omitted part of the forward Bernoulli series has exactly the same or smaller order and is therefore covered by the estimate.

11.4 Remainder of the refined expansion

For

$$H_N^* = 1 + \sum_{n=1}^{N} b_n(L_*) t^{n+1}, \qquad t = Q_*^{-2}, \tag{116}$$

the normalized residual is

$$\mathcal{E}_*(H_N^*; t, L_*) = O(t^{N+2}). \tag{117}$$

After multiplication by $Q_*^2/4$, the forward residual is $O(Q_*^{-2N-2})$. Division by the derivative $\Psi'(Q_*) \sim Q_* L_*$ yields

$$q - Q_* H_N^* = O(Q_*^{-2N-3} L_*^{-1}), \tag{118}$$

which is (72).

Proposition 11.3 (Leading behavior of the refined coefficients). *For every fixed $n \geq 1$,*

$$b_n(L) = -\frac{c_n}{L} + O(L^{-2}), \qquad L \to +\infty. \tag{119}$$

Proof. At gap $2n+2$, the direct contribution of the forward term $c_n q^{-2n}$ to the normalized equation is $4c_n t^{n+1}$. Division by the linearization $4L$ gives $-c_n/L$. Every nonlinear interaction contains at least one previously computed $b_j = O(L^{-1})$, and is therefore $O(L^{-2})$ after the final division by L. □

12 Numerical validation

12.1 Reference evaluation

For high-precision tests it is convenient to avoid direct evaluation of a possibly enormous $\mathrm{K}(x)$. Instead use

$$\log \mathrm{K}(x) = (x-1)\log \Gamma(x) - \log \mathrm{G}(x), \tag{120}$$

and solve $\log \mathrm{K}(x) = Y$. The ordinary and refined anchors provide excellent initial guesses, so one Newton step is usually enough to create a reference value at the displayed precision.

12.2 Error table

Table 2 compares signed errors. In the ordinary columns, $X_0 = Q + 1/2$, $X_1 = X_0 + a_1/Q$, and $X_2 = X_1 + a_2/Q^3$. In the refined columns, $X_0^* = Q_* + 1/2$, $X_1^* = X_0^* + b_1/Q_*^3$, and $X_2^* = X_1^* + b_2/Q_*^5$.

Table 2: Signed approximation errors $X - \mathrm{K}_+^{-1}(e^Y)$.

Y	$X_0 - x$	$X_1 - x$	$X_2 - x$	$X_0^* - x$	$X_1^* - x$	$X_2^* - x$
10	9.162×10^{-3}	1.935×10^{-5}	1.347×10^{-7}	-9.058×10^{-6}	5.793×10^{-8}	-1.632×10^{-9}
100	1.070×10^{-3}	-2.046×10^{-7}	5.165×10^{-10}	-4.585×10^{-7}	5.911×10^{-10}	-2.978×10^{-12}
1000	-1.503×10^{-4}	-2.134×10^{-8}	4.241×10^{-12}	-1.923×10^{-8}	3.898×10^{-12}	-2.917×10^{-15}
10^6	-3.809×10^{-5}	-2.524×10^{-13}	6.128×10^{-19}	-9.592×10^{-13}	4.268×10^{-19}	-6.569×10^{-25}

The data illustrate several structural predictions:

- one ordinary correction improves the anchor by roughly two powers of Q;
- the refined anchor without any tail correction is already comparable to an ordinary expansion with one correction;
- each refined correction again gains approximately two powers of Q_*, with an additional slow factor consistent with L_*^{-1};

- the asymptotic formulas are useful well before Y becomes enormous.

  For reference, the corresponding exact inverse values are

$$\begin{aligned}\mathrm{K}_+^{-1}(e^{10}) &= 4.96615794549670922289\ldots,\\ \mathrm{K}_+^{-1}(e^{100}) &= 10.915136726993084682\ldots,\\ \mathrm{K}_+^{-1}(e^{1000}) &= 27.284589040600087008\ldots,\\ \mathrm{K}_+^{-1}(e^{10^6}) &= 584.23586441169240127\ldots.\end{aligned}$$

Conjecture 12.1 (Enveloping refined expansion)**.** For sufficiently large positive Y, the refined truncations in (12) alternate around the exact inverse and the error after truncation has the sign and approximate magnitude of the first omitted term. Equivalently, the refined series is eventually enveloping on the positive real ray.

The sign pattern in Table 2 and the alternating Bernoulli input support this conjecture, but a proof would require a sign-controlled remainder representation for the centered Hurwitz-zeta derivative and a monotone transfer through the inverse map.

13 Complex branches and exponentially small sectors

13.1 Sectorial Lambert branches

The forward expansion (30) is valid in any closed subsector of the cut plane once a branch of $\log q$ is fixed. The dominant equation $\Phi(Q) = Y$ has sectorial solutions

$$w_k = \mathrm{W}_k\left(\frac{4Y}{\mathrm{e}}\right), \qquad Q_k = \exp\left(\frac{w_k+1}{2}\right), \qquad k \in \mathbb{Z}, \tag{121}$$

provided the branch of $\log Q_k$ is chosen so that $2\log Q_k - 1 = w_k$. The algebraic inverse transseries on that sector is obtained from (6) by replacing (Q, L) with $(Q_k, (w_k+1)/2)$. The coefficient functions themselves do not change.

For the general anchor (79), the branch formula is

$$U_k = \mathrm{W}_k\left(\frac{p}{a}e^{-p\beta}Y\right), \qquad X_k = \exp\left(\beta + \frac{U_k}{p}\right). \tag{122}$$

This expression is safer than writing a free p-th root: it records the compatibility between the logarithm and the Lambert branch. For integer p, different k asymptotically generate the expected root rotations, but the correspondence is modified by the slowly varying logarithmic phase.

13.2 Critical points and branch loss

The anchor derivative vanishes when

$$U + 1 = 0. \tag{123}$$

In the ordinary Lambert coordinate this is the familiar branch point of W at argument $-1/\mathrm{e}$. Near such a point, division by $U + 1$ in (85) is singular and the inverse is no longer described by an ordinary power transseries in the same coordinate. A square-root local model or a uniform coalescing-saddle expansion is required. The large positive K-branch stays far from this obstruction because $U \to +\infty$.

The refined r-Lambert anchor has analogous critical points determined by

$$\mathrm{e}^s(1+s) + r = 0. \tag{124}$$

For $r = -1/(12\mathrm{e})$ and the large positive branch, the derivative is positive and no ambiguity occurs.

13.3 Stokes transfer to the inverse

The algebraic expansion of $\zeta'(-1, q+1/2)$, and equivalently of the Barnes G-function, possesses exponentially improved refinements [11, 12]. Across Stokes rays, exponentially small forward contributions switch on smoothly. Suppose schematically that

$$F(q) = F_{\text{alg}}(q) + \mathfrak{S}(q)e^{-\chi(q)}, \qquad \Re\chi(q) > 0, \tag{125}$$

where F_{alg} is the optimally truncated algebraic expansion. Let q_{alg} solve $F_{\text{alg}}(q_{\text{alg}}) = Y$. The first inverse Stokes sector is then

$$q - q_{\text{alg}} \sim -\frac{\mathfrak{S}(q_{\text{alg}})e^{-\chi(q_{\text{alg}})}}{F'_{\text{alg}}(q_{\text{alg}})}. \tag{126}$$

Higher powers and interactions follow from the same anchored Lagrange–Bürmann formula. Thus inversion does not create a new exponential action at first order; it transports the forward action and divides its amplitude by the logarithmic derivative.

Research status

Equation (126) is a rigorous formal transfer rule once an exponentially improved forward expansion with controlled remainder is supplied. Determining the complete Stokes multipliers of the inverse K-function on all complex sectors is a separate problem and is not claimed here. Existing remainder representations for the Hurwitz zeta derivative and Barnes G provide the natural starting point.

14 Extensions to related special functions

14.1 Generalized hyperfactorial products

For a fixed real $r > -1$, consider the discrete family

$$H_r(n) = \prod_{k=1}^{n} k^{k^r}, \qquad \log H_r(n) = \sum_{k=1}^{n} k^r \log k. \tag{127}$$

Euler–Maclaurin summation gives the dominant term

$$\log H_r(n) \sim \frac{n^{r+1}}{r+1}\left(\log n - \frac{1}{r+1}\right). \tag{128}$$

Put $p = r+1$. The universal anchor is therefore

$$X_r(Y) = \left(\frac{p^2 Y}{\mathrm{W}_0(p^2 Y/\mathrm{e})}\right)^{1/p}. \tag{129}$$

The ordinary hyperfactorial corresponds to $r = 1$, $p = 2$, and reproduces $Q = 2\sqrt{Y/\mathrm{W}_0(4Y/\mathrm{e})}$. Once a chosen analytic continuation is expanded in a translated Euler–Maclaurin normal form, the support-transfer theorem generates its inverse transseries without further conceptual changes.

For integer r, the coefficients are naturally expressed through Bernoulli polynomials and generalized Glaisher–Kinkelin constants. The translation normal form predicts which center should be used before inversion; the center must be derived from the coefficient of the $n^r \log n$ boundary term rather than assumed in advance.

14.2 Inverse gamma and Barnes functions

The inverse gamma and inverse Barnes-G problems are neighboring members of the same class:

$$\log\Gamma(x) = x(\log x - 1) - \frac{1}{2}\log x + \frac{1}{2}\log(2\pi) + O(x^{-1}),$$
$$\log \mathrm{G}(x+1) = \frac{x^2}{2}\left(\log x - \frac{3}{2}\right) - \frac{1}{12}\log x + \zeta'(-1) + O(x^{-2}).$$

The ordinary anchors are respectively

$$X_\Gamma(Y) = \frac{Y}{\mathrm{W}_0(Y/\mathrm{e})}, \tag{130}$$

$$X_G(Y) = 2\sqrt{\frac{Y}{\mathrm{W}_0(4Ye^{-3})}}. \tag{131}$$

The same affine-core theorem can absorb their logarithmic and constant defects into appropriate r-Lambert anchors. The difference among these functions lies primarily in the translated support and coefficient data, not in the inversion mechanism.

14.3 Barnes multiple gamma functions

After a conventional sign normalization, logarithms of Barnes multiple gamma functions [17] have dominant terms of the form $x^p \log x$, with p equal to the order of the relevant multiple-gamma level. Their Euler–Maclaurin tails are governed by multiple Bernoulli polynomials. The general framework of Section 8 therefore predicts:

1. a Lambert anchor with the corresponding exponent p;
2. a translation normal form determined by the first boundary logarithm;
3. an inverse support equal to the additive semigroup generated by the translated multiple-Bernoulli gaps;
4. coefficient functions in a differential algebra generated by the slow Lambert variable and the parameters of the multiple gamma function.

Deriving explicit centered coefficients is largely an algebraic task once the forward multiple-Bernoulli expansion is fixed.

14.4 Parameter-dependent and double-scaling inversions

If a, p, β, or the forward coefficients depend on a parameter θ, the preceding results remain uniform provided

$$|a(\theta)| \geq a_0 > 0, \qquad |U(\theta) + 1| \geq u_0 > 0, \tag{132}$$

and the coefficient algebra has uniform asymptotic bounds. New behavior occurs when $U+1$ approaches zero or when a forward gap tends to zero with θ. Such limits require a double-scaling variable and generally replace the Lambert normal form by a uniform canonical function.

15 Open problems and research directions

15.1 Exponentially improved inverse expansions

Combine a terminant-based or resurgence-based expansion for $\zeta'(-1, z)$ with the anchored formula to compute explicit inverse Stokes multipliers. The leading transfer is (126); the nontrivial work is to organize interactions between the algebraic inverse sectors and the exponentially small forward sectors near Stokes smoothing regions.

15.2 A proof of the enveloping property

Theorem 12.1 should be approachable through a sign-definite integral remainder for the centered expansion. The midpoint shift may be essential: its Bernoulli-polynomial coefficients alternate cleanly, while the unshifted series mixes parity and obscures monotonicity. A successful proof would give rigorous one-sided bounds for K_+^{-1} from successive refined truncations.

15.3 Global real and complex branch census

The two positive stationary points described in Section 2 account for the local real branch changes, but the analytic continuation has a much richer complex critical-value set. A useful program is to locate zeros of

$$\log\Gamma(z) + z - \frac{1}{2} - \frac{1}{2}\log(2\pi) = 0 \tag{133}$$

and map them through K. These critical values determine the branch points and the natural domains of inverse branches. The Lambert branch model (121) gives their large-index asymptotic organization.

15.4 Automatic normal-form selection

A symbolic system could inspect a forward transseries, choose a translation that removes the largest subleading logarithmic monomial, test whether the remaining affine logarithmic core is r-Lambert invertible, and then build the sparse inverse support automatically. The selection criterion in Section 9 suggests a deterministic optimization problem over a small library of exact anchors.

15.5 Formal verification

The algebraic core is well suited to mechanization. A formal development can separate:

1. the exact identities for K, Hurwitz zeta, and Barnes G;
2. finite coefficient identities in Bernoulli polynomials;
3. the formal Lagrange–Bürmann theorem in a complete filtered ring;
4. the support-transfer theorem for finitely generated grids;
5. analytic remainder estimates on the positive ray.

The triangular recurrences are especially friendly to proof assistants because each coefficient theorem reduces to a finite ring identity after the previous orders are available.

15.6 Arithmetic of inverse coefficients

The coefficients c_m have controlled Bernoulli denominators, while $a_n(L)$ and $b_n(L)$ are rational functions with highly structured integer numerators. It is natural to ask for denominator bounds, p-adic valuations, and combinatorial formulas analogous to the Stirling-number formulas for the ordinary Lambert expansion. The operator formula (89) provides a starting point: its repeated first-order operators suggest Bell-polynomial and rooted-tree interpretations.

16 Conclusion

The large inverse of the generalized hyperfactorial is controlled by a quadratic logarithmic saddle, but its clean transseries is visible only after two normalizations. First, passing from x to $q = x - 1/2$ removes the subleading $q \log q$ term and converts the Hurwitz-zeta tail into an even Bernoulli-polynomial series. Second, solving the dominant core exactly with a Lambert function separates nested logarithms from genuinely new inverse-power sectors.

The ordinary anchor

$$Q = 2\sqrt{\frac{\log y}{\mathrm{W}_0(4\log y/\mathrm{e})}}$$

leads to

$$\mathrm{K}_+^{-1}(y) \sim \frac{1}{2} + Q + \frac{a_1}{Q} + \frac{a_2}{Q^3} + \cdots .$$

The affine logarithmic core has an exact r-Lambert inverse and produces the sharper organization

$$\mathrm{K}_+^{-1}(y) \sim \frac{1}{2} + Q_* + \frac{b_1}{Q_*^3} + \frac{b_2}{Q_*^5} + \cdots .$$

Both expansions have explicit all-orders operator formulas, sparse triangular recurrences, and direct residual estimates.

More generally, an anchor of shape $ax^p(\log x - \beta)$ converts the inverse problem into a filtered near-identity problem. Forward power gaps add under nonlinear interaction and transfer to inverse powers by the rule $\sigma \mapsto 1-\sigma$. This support law is the central reusable result: it predicts the architecture of an inverse transseries before any coefficients are computed, applies uniformly to gamma, Barnes, and generalized hyperfactorial families, and provides a natural interface between symbolic algebra, numerical Newton correction, and future formal verification.

A Low-order coefficient audit

This appendix records enough intermediate algebra to make the displayed coefficients independently checkable.

A.1 Forward coefficients

From (32),

m	c_m	contribution to $\log \mathrm{K}(q+1/2)$
1	$-7/5760$	$-7/(5760q^2)$
2	$31/161280$	$31/(161280q^4)$
3	$-127/1290240$	$-127/(1290240q^6)$
4	$511/4866048$	$511/(4866048q^8)$
5	$-1414477/7380172800$	$-1414477/(7380172800q^{10})$

A.2 Ordinary recurrence at the first orders

Write

$$H = 1 + a_1 t + a_2 t^2 + a_3 t^3 + O(t^4).$$

Expanding (41) gives

$$[t]\mathcal{E} = 4(La_1 - \Delta), \tag{134}$$

$$[t^2]\mathcal{E} = 2La_1^2 + 4La_2 + 2a_1^2 - \frac{a_1}{6} - \frac{7}{1440}, \tag{135}$$

$$[t^3]\mathcal{E} = 4La_1a_2 + 4La_3 + \frac{2}{3}a_1^3 + \frac{1}{12}a_1^2 + 4a_1a_2 + \frac{7}{720}a_1 - \frac{1}{6}a_2 + \frac{31}{40320}. \tag{136}$$

The first equation gives $a_1 = \Delta/L$. Substitution into the second gives (8); substitution of both into the third gives (9). These equations are also useful unit tests for a symbolic implementation.

A.3 Refined recurrence at the first orders

Write

$$H = 1 + b_1 t^2 + b_2 t^3 + b_3 t^4 + O(t^5).$$

Expansion of (65) gives

$$[t^2]\mathcal{E}_* = 4Lb_1 - \frac{7}{1440}, \tag{137}$$

$$[t^3]\mathcal{E}_* = 4Lb_2 - \frac{1}{6}b_1 + \frac{31}{40320}, \tag{138}$$

$$[t^4]\mathcal{E}_* = 2Lb_1^2 + 4Lb_3 + 2b_1^2 + \frac{7}{720}b_1 - \frac{1}{6}b_2 - \frac{127}{322560}. \tag{139}$$

Solving successively reproduces (13)–(15).

A.4 Independent first-order check

The first anchored correction is $-R(Q)/\Phi'(Q)$. Since

$$R(Q) = \kappa - \frac{L}{24} + \frac{c_1}{Q^2} + O(Q^{-4}) = -\Delta + \frac{c_1}{Q^2} + O(Q^{-4}),$$

we obtain

$$-\frac{R(Q)}{QL} = \frac{\Delta}{LQ} - \frac{c_1}{LQ^3} + O(Q^{-5}),$$

which verifies a_1 and the direct Bernoulli contribution $7/(5760L)$ inside a_2. The remaining pieces of a_2 are the quadratic terms in (51).

B Reference implementation

The following self-contained Python code uses `mpmath`. It accepts $Y = \log y$ rather than y, avoiding overflow when the original argument is enormous. The functions `ordinary` and `refined` implement the first three correction blocks displayed in the article; `polish` uses the exact logarithmic derivative for optional Newton refinement.

```
import mpmath as mp

mp.mp.dps = 80
KAPPA = mp.log(mp.glaisher) - mp.mpf(1) / 8


def log_k(x):
    """log K(x), evaluated through Gamma and Barnes G."""
    x = mp.mpf(x)
    return (x - 1) * mp.loggamma(x) - mp.log(mp.barnesg(x))


def d_log_k(x):
    """Exact derivative of log K(x)."""
    x = mp.mpf(x)
    return (mp.loggamma(x) + x - mp.mpf('0.5')
            - mp.log(2 * mp.pi) / 2)


def ordinary_coefficients(L):
    """a_1, a_2, a_3 for the ordinary Lambert anchor."""
```

```
    D = L / 24 - KAPPA
    a1 = D / L
    a2 = (mp.mpf(7) / (5760 * L)
          + D / (24 * L**2)
          - (L + 1) * D**2 / (2 * L**3))
    a3 = ((3 * L**2 + 5 * L + 3) * D**3 / (6 * L**5)
          - (4 * L + 3) * D**2 / (48 * L**4)
          + (1 - 7 * L) * D / (1920 * L**3)
          + (49 - 186 * L) / (967680 * L**2))
    return [a1, a2, a3]


def ordinary(Y, terms=3):
    """Asymptotic inverse of log K(x)=Y, ordinary anchor."""
    Y = mp.mpf(Y)
    if not 0 <= terms <= 3:
        raise ValueError("terms must be between 0 and 3")
    w = mp.lambertw(4 * Y / mp.e)
    Q = 2 * mp.sqrt(Y / w)
    L = mp.log(Q)
    x = Q + mp.mpf('0.5')
    for n, a in enumerate(ordinary_coefficients(L)[:terms], start=1):
        x += a / Q**(2 * n - 1)
    return mp.re(x)


def r_lambert_positive(z, r, max_steps=30):
    """Positive large solution of s*exp(s)+r*s=z by Newton iteration."""
    z = mp.mpf(z)
    r = mp.mpf(r)
    s = mp.lambertw(z)  # excellent initial value when z is large
    for _ in range(max_steps):
        es = mp.exp(s)
        f = s * es + r * s - z
        fp = es * (s + 1) + r
        step = f / fp
        s -= step
        if abs(step) <= mp.eps * max(1, abs(s)):
            break
    return s


def refined_coefficients(L):
    """b_1, b_2, b_3 for the refined r-Lambert anchor."""
    b1 = mp.mpf(7) / (5760 * L)
    b2 = -(186 * L - 49) / (967680 * L**2)
    b3 = ((45720 * L**2 - 5435 * L + 637)
          / (464486400 * L**3))
    return [b1, b2, b3]


def refined(Y, terms=3):
    """Asymptotic inverse of log K(x)=Y, refined r-Lambert anchor."""
    Y = mp.mpf(Y)
    if not 0 <= terms <= 3:
        raise ValueError("terms must be between 0 and 3")
    r = -1 / (12 * mp.e)
    z = (48 * (Y - KAPPA) + 1) / (12 * mp.e)
```

```
    s = r_lambert_positive(z, r)
    Q = mp.exp((s + 1) / 2)
    L = mp.log(Q)
    x = Q + mp.mpf('0.5')
    for n, b in enumerate(refined_coefficients(L)[:terms], start=1):
        x += b / Q**(2 * n + 1)
    return x


def polish(Y, x, steps=1):
    """Newton polishing using exact log K and its exact derivative."""
    Y = mp.mpf(Y)
    x = mp.mpf(x)
    for _ in range(steps):
        x -= (log_k(x) - Y) / d_log_k(x)
    return x


if __name__ == "__main__":
    for Y in [10, 100, 1000, mp.mpf(10)**6]:
        x0 = refined(Y, terms=2)
        x = polish(Y, x0, steps=2)
        print("Y =", Y)
        print("asymptotic =", mp.nstr(x0, 30))
        print("polished   =", mp.nstr(x, 30))
        print("residual   =", mp.nstr(log_k(x0) - Y, 8))
```

C Notation and branch conventions

Symbol	Meaning
$\mathrm{K}(z)$	Generalized hyperfactorial normalized by $\mathrm{K}(z+1) = z^z\,\mathrm{K}(z)$, $\mathrm{K}(1) = 1$.
K_+^{-1}	Large positive real inverse branch.
$\mathrm{G}(z)$	Barnes G-function.
A	Glaisher–Kinkelin constant.
Y	$\log y$, the logarithm of the argument of the inverse.
q	Centered variable $x - 1/2$.
w	$\mathrm{W}_0(4Y/\mathrm{e})$.
Q	Ordinary Lambert anchor $\exp((w+1)/2)$.
L	$\log Q = (w+1)/2$.
Q_*, L_*	Refined r-Lambert anchor and its logarithm.
$\mathcal{W}_r$	Inverse of $s \mapsto se^s + rs$.
$B_n(x)$	Bernoulli polynomial; $B_n = B_n(0)$.
Λ	Set of positive forward power gaps.
$\langle\Lambda\rangle$	Additive semigroup generated by Λ.
$\mathcal{D}_X$	Anchor derivative $(F_0'(X))^{-1}\,\mathrm{d}/\,\mathrm{d}X$.

Acknowledgment of computational checks

The rational coefficient identities and the numerical table were independently checked by symbolic series expansion and high-precision numerical evaluation of (120). These checks are not substitutes for the proofs; they serve as regression tests for signs, shifts, and denominator arithmetic.

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